

Controlling Eye Blink for Talking Face Generation via Eye Conversion

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ABSTRACT

A real talking face video includes not only the movement of the mouth, but also realistic blinking details. For a computer generated talking face video, realistic eye movements are critical to overcome the uncanny valley effect. However, it remains a great challenge to introduce realistic eye movements into talking face generation systems. In this paper, we propose a two-stage system for generating talking face video with realistic controllable blinking actions. Through eye conversion and frame replacement, our architecture can ensure the controllability of the blinking motion generation. We propose an eye conversion GAN, which can convert a face image into any stages of blinking, and maintain the consistency of facial identity features. In this network, we design joint training to increase the network's ability of generating closed and half-closed eye images, which improves the authenticity of the eyes. Experiments on two popular data sets show that compared with previous work, our method can not only guarantee the authenticity of mouth movements, but also generate realistic and controllable eye blinks.

CCS CONCEPTS

• **Computing methodologies** → **Animation; Image processing.**

KEYWORDS

talking face generation, eye blink generation, eye conversion

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1 INTRODUCTION

People usually blink when speaking, and this subtle facial movement is also a necessary element to make a dynamic face vivid. Therefore, realistic eye blinking is essential for talking face generation. In the traditional speech-driven talking face generation

work, by inputting an image of a target face and a piece of audio, the architecture synthesizes a video in which the mouth movement matches the input audio. However, the eye blinking is usually lacking, which greatly impairs the authenticity of the generated video. Due to the uncanny valley effect, realistic mouth movements with unnatural eye movements can make people feel uncomfortable, and this feeling is especially strong when the blinking movements are fixed or uncontrollable. Therefore, ensuring the faithfulness of the eye movements in the generated results is critical for improving the authenticity of the talking face video.

At the earliest, talking face generation focused on improving the synchronization of mouth movements and speech. These methods include methods for specific talking face generation [Suwajanakorn et al. 2017; Wang et al. 2013; Xie and Liu 2007] and other architectures that can be extended to any face [Chung et al. 2017; Liu et al. 2015; Taylor et al. 2017; Yu et al. 2021; Zhou et al. 2019]. Averbuch-Elor et al. [2017] provide a way to manipulate facial expressions. However, they all lack the ability to generate eye details.

To this end, Vougioukas et al. [2019] introduced random noise to drive blinks. Although the frequency of blinking can be constrained to some certain extent by setting noise generation rules, the correlation between blinks and voice scenarios is weakened, and it is difficult to control the generation of blinks. Zhou et al. [2020] uses landmarks to guide the generation of blinking movements. However, the actual blinking action is neither unfixed frequency and immutable, nor completely random. It is related to the speaker's habits and the content of the speech. Therefore, a fixed blink generation mode is still unsuitable to cover various situations to generate sufficiently realistic blink actions. If a network simply learns the relationship between blinking and mouth movements, the blinking pattern learned by the model will be not realistic enough.

Talking face generation requires not only the authenticity and matching ability of blinking, but also the scalability of faithful generation effect. This means that a reliable talking face generation method needs to have the ability of controllable blink generation. Specifically, within a same input audio, the architecture must be able to generate not only a blinking effect consistent with the ground truth, but also any blinking action pattern whenever the user needs it (e.g., long-time closed eyes), aiming to match different one's habit or to achieve entertainment effects. In this case, learning the association between blinks and speech through a single model cannot meet the goal of controllability and arbitrariness of blink generation. Therefore, we propose a two-stage system to introduce controllable blinking actions when generating a talking face video, so that for any input voice, the blinking pattern can be freely controlled to match different persons' habits or achieve different effects.

Figure 1 illustrates the architecture of our system. In the first stage of the pipeline, we build a general structure for speech driven

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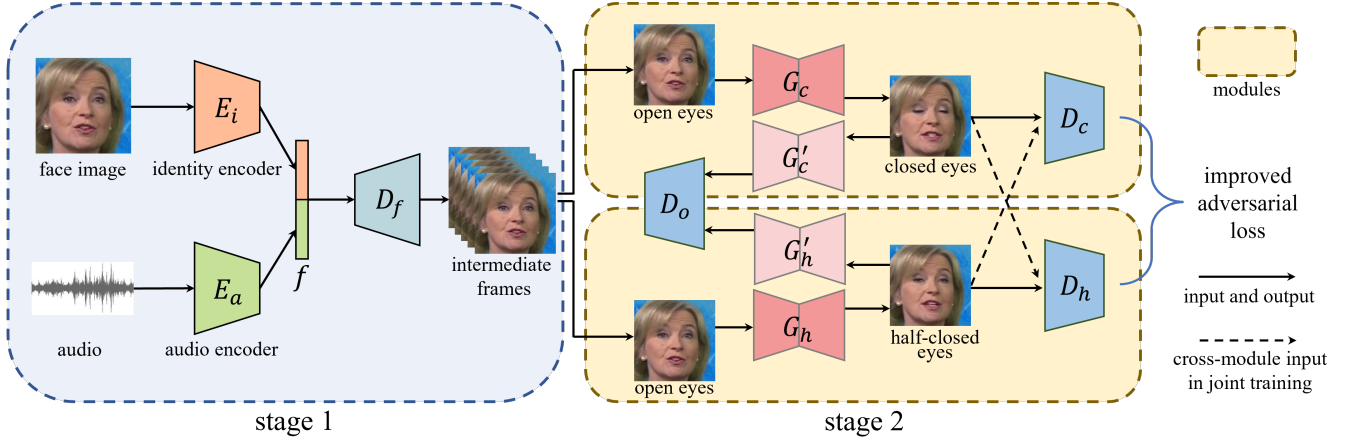


Figure 1: The architecture of the proposed system. Face images: © BBC.

talking face generation. We use two encoders (E_i and E_a in Fig. 1) to extract the identity feature and the audio feature. Then we use a decoder (D_f) to accept the two connected features to produce the frames without eye blink. In the second stage, we propose a jointly trained GAN to replace the original frame by generating the half-closed and closed-eye face images required for regular eye blinking, thereby introducing the blinking action into the original video. We propose to use joint training to help optimize its eye conversion ability and overcome the shortcomings of the half-closed eye data set. Since the GAN can process video frames independently, and the choice of frame replacement can also be independently specified, the blinking action and pattern are completely controllable, which guarantees the authenticity of the resulting blinking pattern.

2 APPROACH

2.1 Synthesis of the Mouth Movements

At this stage, we first generate mouth movements. This requires that the mouth motion is synchronized with speech while ensuring that the identity of the input face remains unchanged. We build an identity feature encoder E_i (orange module in Fig. 1) to encode the input face image into identity features. Similarly, a speech encoder E_a (green module in Fig. 1) is used to convert audio clips into speech features. After the two features are connected, the joint feature f is fed into the decoder D_f to generate face image with mouth shape synchronized with the speech.

In order to dissociate the relationship between speech and eye movements to prevent the erroneous blinking movements in the first stage, we adapt the method of Li et al. [2018] to perform blink detection on the training data, and remove images with half-closed or closed eyes in the data set. In DAVS [Zhou et al. 2019], Zhou et al. disentangled identity and audio features, which enables their two encoders to better extract corresponding features without interfering with each other. This can help the decoder to better learn the mapping of sounds to mouth movements while preserving facial features. Therefore, we use the DAVS model as a pre-trained model, and train the model with the filtered data set. We add the $L1$ loss to

optimize the decoder's retention of input identity features:

$$\mathcal{L}_{id} = \|I_{id} - D_f(f)\|_1 \quad (1)$$

where I_{id} is the face image that is fed into the identity encoder.

Similarly, in order to strengthen the synchronization ability between the mouth and speech, we build the mouth loss with mask:

$$\mathcal{L}_{mouth} = \|S \cdot M - D_f(f) \cdot M\|_1 \quad (2)$$

where S is the real frame, and M is the mask which highlights the mouth area. Therefore, the overall loss is:

$$\mathcal{L}_1 = \mathcal{L}_{id} + \lambda \mathcal{L}_{mouth} \quad (3)$$

where λ is the coefficient to balance the weights of each loss.

2.2 Synthesis of the Eyes

When observing multiple videos of the LRW data set, we found that one's blinking action usually lasts for 3-4 frames at a frame rate of 25-30 frames per second, indicating that a complete blink motion requires frames with half-closed, closed, and half-closed eyes in order. Therefore, to synthesize a complete blinking action, we only need to convert the consecutive original frames into new frames with half-closed and closed eyes sequentially. In addition, these two states can also imitate other eye modes, such as the state of thinking with eyes closed for a long time. Based on this observation, we propose an eye conversion GAN network (stage 2 in Fig. 1), to generate images with different eye states, and produce the process of blinking through frame replacement.

As mentioned above, the network needs to have the ability to convert two domains, so our eye conversion GAN network contains two parallel modules, namely closed eye module and half-closed eye module. Each module consists of two generators: G_c , G'_c , G_h and G'_h . We devise the corresponding discriminator D_c and D_h in each module to distinguish closed and half-closed eyes, and set D_o to identify the open-eye images reconstructed from G'_c and G'_h . Since the changes of the eyelids between the two eye states are small enough, the smoothness of blinking can be naturally guaranteed by two modules in a video with 25-30 frames per second.

Table 1: Quantitative Evaluation Results

Method	VoxCeleb2				LRW			
	Full image		Eye area		Full image		Eye area	
	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
DAVS-speech driven	21.45	0.8176	21.13	0.7471	23.75	0.8432	23.06	0.8083
DAVS-video driven	21.96	0.8327	21.31	0.7602	25.38	0.8837	24.82	0.8515
Vougioukas et al.	20.86	0.7202	21.20	0.7730	20.39	0.7703	23.94	0.8004
Proposed model	22.05	0.8358	22.17	0.7855	24.01	0.8537	24.99	0.8520

Initially, we use two independent cycleGAN [Zhu et al. 2017] networks to train the closed-eye and half-closed eye conversion modules, respectively. However, due to the inconsistent degree of closed eyes in the half-closed eye data set, the conversion results of the half-closed eye module are poor. Fortunately, we found that in the early stage of the closed-eye module training, the image output by the generator has a similar effect on the half-closed eyes to some extent. Therefore, in order to overcome the training difficulties by half-closed eyes, we conduct joint training on the two modules. Note that the two modules use the closed-eye data set for training and share parameters in the early stage of training and we train with their respective data sets afterwards.

In order to make the two modules make full use of each other's training data, and avoid flickering in consecutive blinking frames, we improve the adversarial loss [Goodfellow et al. 2014] by making the discriminators of the two modules cooperate with each other. While improving the training effect of the half-closed eye module, it also increases the ability of the two modules to transform the eyes. Taking the generator G_h as an example, the improved adversarial loss is expressed as:

$$\mathcal{L}'_{\text{GAN}}(G_h, D'_h) = \mathbb{E}[\log D'_h(h)] + \mathbb{E}[\log(1 - D'_h(G_h(o)))] \quad (4)$$

$$D'_h(x) = \alpha D_h(x) + (1 - \alpha) D_c(x) \quad (5)$$

where O is the open eye domain, and the H is the half-closed eye domain. The lowercase letters o and h represent the images of the corresponding domain, respectively. The x represents any image which is fed into the discriminator. The α is set to 1 at first, and it gradually decreases as the training progresses.

For the generators G'_c and G'_h that convert from closed and half-closed eye domains to open eye domains, the two modules will share the same discriminator D_o . In the process of optimizing the discriminator, the images in the two domains are considered equivalently, written as:

$$\begin{aligned} \max_{D_o} \mathcal{L}'_{\text{GAN}}(D_o) &= \mathbb{E}[\log D_o(o)] + \frac{1}{2} \mathbb{E}[\log(1 - D_o(G'_c(c)))] \\ &\quad + \frac{1}{2} \mathbb{E}[\log(1 - D_o(G'_h(h)))] \end{aligned} \quad (6)$$

where c means the images with closed eyes.

We use the identity loss and cycle consistency loss similar to cycleGAN to improve the authenticity and cycle consistency of the input image. Take module A as an example, the identity loss and the cycle consistency loss can be expressed as:

$$\mathcal{L}_{\text{identity}}(G_c, G'_o) = \|G_c(c) - c\|_1 + \|G'_c(o) - o\|_1 \quad (7)$$

$$\mathcal{L}_{\text{cycle}}(G_c, G'_c) = \|G'_c(G_c(o)) - o\|_1 + \|G_c(G'_c(c)) - c\|_1 \quad (8)$$

As a result, the full objective is:

$$\mathcal{L}_2 = \mathcal{L}'_{\text{GAN}} + \mu \mathcal{L}_{\text{cycle}} + \nu \mathcal{L}_{\text{identity}} \quad (9)$$

where μ and ν are the coefficients to control the weights. We set μ and ν to 5 and 10, respectively.

3 EXPERIMENTS

The data set for training the eye conversion GAN is generated based on the LRW data set [Chung and Zisserman 2017]. We first randomly sample the frames in the LRW data set, and use the blink detection method [Li et al. 2018] to process each frame to obtain a score of the possibility of closed eyes. According to the score, each frame is classified as an image with open eyes, closed eyes, or half-closed eyes. Then these frames are added to the corresponding training set. Finally, we obtained 80,000 open-eye images, 61,218 closed-eye images, and 41,276 half-closed-eye images as the training set, and the test set includes 2,500 images in each domain.

We implemented our network with PyTorch and trained it on a computer equipped with a 2080Ti GPU for about 1 week. When we began to train the eye conversion GAN, we only used the closed eye data set (including unpaired face images with open eyes and closed eyes) to train the closed eye module. Then we copied the closed eye module to the half-closed eye module and began to use their respective data sets to separately train the two modules.

We quantitatively evaluated our method on the LRW [Chung and Zisserman 2017] and VoxCeleb2 [Chung et al. 2018] data sets, and compared with the methods of DAVS [Zhou et al. 2019] and [Vougioukas et al. 2019]'s method. We used PSNR and SSIM to evaluate the similarity between the generated video and ground truth, and the comparison results are shown in Table 1. All methods used the same identity images and audio as input, and we selected the frames that need to be blinked and performed blink conversion processing to generate our the final results. For each frame, we separately compared the eye area and the entire image to evaluate the improvement brought by our method in the eye area and the overall video. The best result of each comparison group is indicated in bold font. The results show that in most cases, our method can improve the authenticity of the video by adding controllable blinking actions. Since our method is speech-driven, there may be a gap in the authenticity of the mouth compared to the video-driven DAVS. Nevertheless, our method achieved better scores than the state-of-the-art methods in terms of the eyes.

Figure 2 shows the results with and without eye blink. It can be seen that our system can generate videos with blinking actions

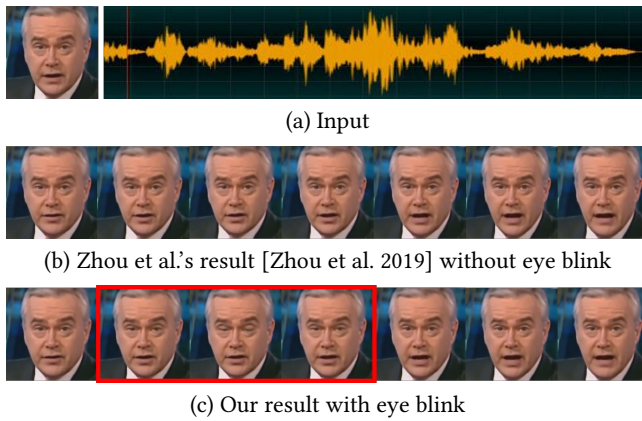


Figure 2: The results with and without eye blink. (a) Input face image: © BBC.

from face image and audio, while maintaining the authenticity of both mouth and eyes.

Figure 3 demonstrates the controllability of our method on blinking effect. By changing the selected frames, we can make blinking or closed eyes occur anywhere in the video. Since our method can control blinking, it can not only generate a video with blinks consistent with ground truth, but also yield blinking patterns that conform to any special situation or adapt to the habits of different people. For example, we can simulate a situation where the eyes are closed continuously (Figure 3(c)). The results show that our method does not damage the synchronization or authenticity of the mouth while ensuring controllable blinking.

4 CONSLUSION

In this work, we proposed a novel architecture for generating a talking face with realistic and controllable blinking. We built a two-stage pipeline to decouple the blinking action from the sound and perform independent control, so that the generation of the blinking action is controllable and has a wider range of applications. In this architecture, we proposed an eye conversion GAN, which used joint training to avoid the difficulty of training for half-closed-eye conversion, and strengthened the conversion effect between the three domains. Various results on multiple data sets showed that we can achieve more realistic results compared with the state-of-art methods. Limited by the data set, our method cannot well handle cartoon faces, especially for characters whose eye ratio is vastly different from the real face. We will extend our method to deal with cartoon faces in the future.

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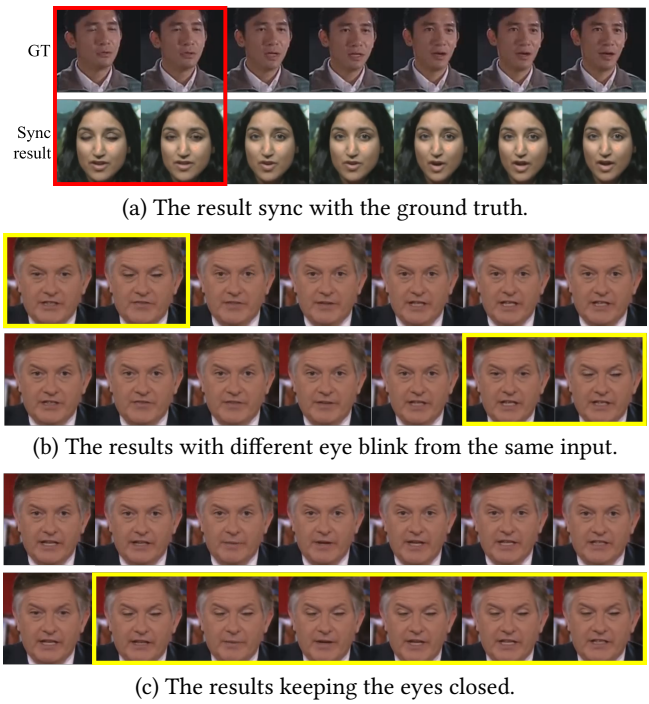


Figure 3: The results by controlling the eyes. All input face images: © BBC.

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